

Supplementary Materials

Description of Standardized Measures Used to Characterize the Autistic Sample

Novel Word Learning from Social Robots by Autistic and Non-Autistic Children

The following standardized instruments were administered to characterize the autistic sample and are summarized in Tables 1 and 2 of the main manuscript. They were used for sample description rather than as eligibility criteria. All assessments were conducted by appropriately trained personnel within the early intervention service. Each measure carried some missingness in the autistic group; the number of children with valid scores is reported in the table notes. Non-autistic children did not complete the diagnostic or developmental assessments, and the corresponding cells are therefore left blank.

Autism Diagnostic Observation Schedule, Second Edition (ADOS-2)

The ADOS-2 (Lord et al., 2012) is a semi-structured, standardized assessment of communication, reciprocal social interaction, play, and restricted and repetitive behaviors. The module administered is selected according to the child's expressive language level, and a calibrated comparison score (range 1 to 10) is derived to index the level of autism-related behaviors relative to same-age, same-language peers, with higher scores indicating a greater degree of observed autism characteristics. In the present study, the ADOS-2 comparison score is reported as a descriptor of the autistic sample.

Mullen Scales of Early Learning (MSEL)

The MSEL (Mullen, 1995) is a standardized measure of cognitive development for children from birth to 68 months, comprising five scales: Gross Motor, Visual Reception, Fine Motor, Receptive Language, and Expressive Language. Because standardized scores show floor effects for a substantial proportion of autistic children, developmental quotients (DQs) were derived from age-equivalent scores to provide a more sensitive index of ability. A developmental quotient was calculated as the age-equivalent score divided by the child's chronological age, multiplied by 100. A Verbal DQ was derived from the Receptive and Expressive Language scales, and a Non-verbal DQ from the Visual Reception and Fine Motor scales. Age-equivalent values were obtained from the published Mullen age-equivalence tables.

Vineland Adaptive Behavior Scales, Third Edition (Vineland-3)

The Vineland-3 (Sparrow et al., 2016) is a caregiver-reported, standardized measure of adaptive functioning across the domains of Communication, Daily Living Skills, and Socialization, with an optional Motor domain. The Adaptive Behavior Composite (ABC) is a norm-referenced standard score (population mean = 100, standard deviation = 15), with lower scores indicating greater adaptive difficulty. In the present study, the ABC is reported as an index of overall adaptive functioning in the autistic sample.

Derivation and reporting notes

Developmental quotients are reported in preference to standardized T-scores or index scores for the MSEL because of well-documented floor effects in this population; this approach retains meaningful between-child variation that standardized scores can obscure. All scores are reported at the group level as means and standard deviations, with per-measure missingness noted, and no individual-level scores are presented.

References

- Lord, C., Rutter, M., DiLavore, P. C., Risi, S., Gotham, K., & Bishop, S. L. (2012). *Autism Diagnostic Observation Schedule, Second Edition (ADOS-2)*. Western Psychological Services.
- Mullen, E. M. (1995). *Mullen Scales of Early Learning (AGS ed.)*. American Guidance Service.
- Sparrow, S. S., Cicchetti, D. V., & Saulnier, C. A. (2016). *Vineland Adaptive Behavior Scales, Third Edition (Vineland-3)*. Pearson.